

Tools

Saved Mar 12, 2026 at 12:17 PM

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Investigating pedagogical approaches for teaching quantitative skills to non-math majors

ExpertReview score

Consent

1 Question

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Confidentiality

Confidentiality

To maintain anonymity while allowing researchers to track your responses over time, please visit [this link](#) and enter your UF ID number to generate a unique participant identification code.

Participant ID

What is your unique participant ID generated using [this link](#)?

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Technology Access

Current classes

Which of the following courses using R are you taking **this semester**?

- ☐ Computational Problem Solving in Wildlife Ecology Using R (WIS 4934)
- ☐ Introduction to Wildlife Population Ecology (WIS 4501)
- ☐ Quantitative Wildlife Ecology (WIS 4601)
- ☐ Other
-

## Previous classes



Which of the following courses using R have you taken **previously**?

- ☐ Computational Problem Solving in Wildlife Ecology Using R (WIS 4934)
- ☐ Introduction to Wildlife Population Ecology (WIS 4501)
- ☐ Quantitative Wildlife Ecology (WIS 4601)
- ☐ Other

## OS

What type of operating system (OS) did you primarily use for this class?

- ☐ Mac
- ☐ Windows
- ☐ Linux
- ☐ Other

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## Computer age

Roughly how old is the computer you primarily used for this class?

- ☐ 0 - 2 years old
- ☐ 2 - 4 years old
- ☐ 4 - 6 years old
- ☐ 6+ years old

## Laptop issues

During this past semester, how often did problems with your primary laptop (e.g., battery life, slowness, connectivity issues, etc.) negatively impact your performance in this class?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Always



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Qualitative assessment

EssayProduceAGraph



Providing as much detail as you can, please describe how you would produce a graph for a lab report or presentation using whichever program or software you prefer if asked to do so today.

EssayLearningR



What challenges do you feel stand in your way for learning R?

EssayRBenefits



What benefits do you see coming from using R or other programming languages?

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▼ R Proficiency

R use

Answer the following questions about your experiences with R *before and after* this semester.

|                                                   | Before this semester  |                       |                       |                       |                       | After this semester   |                       |                       |                       |                       |
|---------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                   | Very unlikely         | Somewhat unlikely     | Neutral               | Somewhat likely       | Very likely           | Very unlikely         | Somewhat unlikely     | Neutral               | Somewhat likely       | Very likely           |
| How likely would you be to use R to analyze data? | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Anxiety

|                                                                                         | Before this semester  |                       |                       |                       |                       | After this semester   |                       |                       |                       |                       |
|-----------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                                         | Very low anxiety      | Low anxiety           | Moderate anxiety      | High anxiety          | Very high anxiety     | Very low anxiety      | Low anxiety           | Moderate anxiety      | High anxiety          | Very high anxiety     |
| How would you rate your anxiety surrounding computer programming tasks (e.g., using R)? | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| How would you rate your anxiety surrounding mathematical concepts and problems?         | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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Computing programs

How would you rate your overall proficiency in the following programs?

|                                                                                                     | Before this semester  |                       |                       |                       |                       | After this semester   |                       |                       |                       |                       |
|-----------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                                                     | No proficiency        | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  | No proficiency        | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  |
| R                                                                                                   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Spreadsheets (e.g., Excel, GoogleSheets)                                                            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Other coding languages (e.g., Python, C++, SAS, MATLAB, etc.)                                       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Graphical user interface-based (GUI) statistical software programs (e.g., JMP, Minitab, SPSS, etc.) | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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R skills

How would you rate your proficiency in the following skills in R?

|                               | <u>Before</u> this semester |                       |                       |                       |                       | <u>After</u> this semester |                       |                       |                       |                       |
|-------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                               | No proficiency              | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  | No proficiency             | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  |
| Importing data                | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Formatting data               | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Plotting a figure             | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Creating a for-loop           | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Creating a function           | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Using external packages in R  | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Debugging and troubleshooting | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Annotating/documenting code   | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Data summarization            | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| RMarkdown                     | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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Math skills

How would you rate your proficiency in the following quantitative skills?

|                                                                           | <u>Before</u> this semester |                       |                       |                       |                       | <u>After</u> this semester |                       |                       |                       |                       |
|---------------------------------------------------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                           | No proficiency              | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  | No proficiency             | Slightly proficient   | Moderately proficient | Very proficient       | Extremely proficient  |
| Hypothesis testing and experimental design                                | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Calculating measures of central tendency (e.g., mean, median) or variance | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Using statistical tests to compare means (e.g., ANOVA, t-test)            | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Linear regression analysis                                                | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Model output (parameter) interpretation                                   | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Comparing fit among models                                                | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Exploratory data analysis                                                 | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Relating ecological/biological questions to quantitative models           | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

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Comfort with R

External help



Answer the following questions about your level of ease in finding external help:

|                                                                                                                     | Not at all comfortable | Slightly comfortable  | Moderately comfortable | Very comfortable      | Extremely comfortable |
|---------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|------------------------|-----------------------|-----------------------|
| How comfortable do you feel with finding help from your classmates with R?                                          | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/> |
| How comfortable do you feel with finding help from the internet (e.g., YouTube videos, StackExchange, etc.) with R? | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/> |
| How comfortable do you feel using official R documentation (e.g., R help files, vignettes)?                         | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/>  | <input type="radio"/> | <input type="radio"/> |

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Perseverance

In general, when faced with an issue in R, how long do you spend working on it before you give up trying to deal with it?

- ☐ 0 - 5 minutes
- ☐ 5 - 30 minutes
- ☐ 30 minutes - 1 hour
- ☐ 1 - 2 hours
- ☐ 2 - 7 hours
- ☐ 7+ hours

Computer savviness

In general, how would you rate your level of technological proficiency?

- ☐ Not technologically proficient
- ☐ Slightly technologically proficient
- ☐ Moderately technologically proficient
- ☐ Very technologically proficient
- ☐ Extremely technologically proficient

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▼ Engagement

Effort

How much effort was required to accomplish your level of performance in this course?

- ☐ No effort
- ☐ Low effort
- ☐ Moderate effort
- ☐ High effort
- ☐ Very high effort

Engagement

In your opinion, in a typical week of class, how engaged were you in the course (e.g., watching/reading required material before class, attending lectures/labs, completing assignments, etc.)?

- ☐ Not engaged
- ☐ Slightly engaged
- ☐ Moderately engaged
- ☐ Very engaged
- ☐ Extremely engaged

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▼ Applicability

Future classes



Answer the following question about your interest in future quantitative courses:

|                                                                                   | <u>Before</u> this semester |                       |                       |                       |                       | <u>After</u> this semester |                       |                       |                       |                       |
|-----------------------------------------------------------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                                                   | Not interested              | Slightly interested   | Moderately interested | Very interested       | Extremely interested  | Not interested             | Slightly interested   | Moderately interested | Very interested       | Extremely interested  |
| How interested are you in taking classes with quantitative content in the future? | <input type="radio"/>       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Job search

To what degree do you think your R coding skills will be helpful in your future job search?

- ☐ Not at all helpful
- ☐ Slightly helpful
- ☐ Moderately helpful
- ☐ Very helpful
- ☐ Extremely helpful

Dream job

How interested are you in a career that requires computational skills (i.e. uses R, Python, etc. regularly)?

- ☐ Not interested
- ☐ Slightly interested
- ☐ Moderately interested
- ☐ Very interested
- ☐ Extremely interested

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Demographics



Age



How old are you?

Gender

How do you describe yourself?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Other
- ☐ Prefer not to say

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Race



Choose one or more races that you consider yourself to be

- ☐ White or Caucasian
- ☐ Black or African American
- ☐ American Indian/Native American or Alaska Native
- ☐ Asian
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other
- ☐ Prefer not to say

Ethnicity



Are you of Spanish, Hispanic, or Latino origin?

- ☐ Yes
- ☐ No

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▼ Academic demographics

Major

What is your major?

- Academic standing
- What is your current academic standing?
- ☐ Freshman

☐ Sophomore

☐ Junior

☐ Senior

☐ Other

☐ Dual undergraduate/graduate degree program

☐ Graduate degree program

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Categories

Indicate whether you fall into any of the following categories:

|                          | Yes                   | No                    |
|--------------------------|-----------------------|-----------------------|
| Transfer student         | <input type="radio"/> | <input type="radio"/> |
| Nontraditional student   | <input type="radio"/> | <input type="radio"/> |
| First generation student | <input type="radio"/> | <input type="radio"/> |
| Minority in STEM         | <input type="radio"/> | <input type="radio"/> |

- High school math
- Did your high school offer computer science/programming or advanced mathematics classes?
- ☐ Yes

☐ No

☐ I'm not sure